MVP Simulator: MDP Formulation and Initial Results

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1 MDP Formulation

The MVP simulator implements a finite-horizon MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$:

1.1 State Space

$$\mathcal{S} = \{\texttt{sedentary},\ \texttt{active}\} \times \mathcal{T}$$

where $\mathcal{T} = \{\texttt{morning}, \texttt{midday}, \texttt{afternoon}, \texttt{evening}, \texttt{night}\}$ are the five daily decision points.

1.2 Action Space

$$\mathcal{A} = \{\texttt{send},\ \texttt{don\_t\_send}\}$$

A binary intervention: send a motivational message or do not.

1.3 Transition Function

$P(s' \mid s, a)$ is defined by a configurable matrix. The probabilities encode a Reactance effect: reminders help sedentary users but can backfire for users who are already active.

Current state s	Action a	$P(\texttt{active})$	$P(\texttt{sedentary})$
sedentary	send	0.3	0.7
sedentary	don't send	0.1	0.9
active	send	0.5	0.5
active	don't send	0.6	0.4

Table 1: Transition probabilities. Sending a reminder increases the probability of becoming active for sedentary users ($0.3 > 0.1$) but decreases it for active users ($0.5 < 0.6$), consistent with Reactance theory.

Time-of-day mask: at `night`, all transitions are suppressed (availability indicator per HeartSteps V2).

1.4 Reward

$$r(s, a) = \begin{cases} 1.0 & \text{if } s'.\texttt{activity} = \texttt{active} \\ 0.0 & \text{otherwise} \end{cases}$$

Single-timescale placeholder. Multi-timescale reward (immediate + delayed body measure) is deferred to Phase 2.

1.5 Episode Structure

$T = 90$ days $\times$ 5 steps/day = 450 timesteps.

2 Agents

All agents are contextual bandits: state is accepted but not used in action selection. For the stationary MVP transition matrix, $E[r \mid a]$ is the relevant quantity.

2.1 Thompson Sampling

Beta-Bernoulli TS with prior Beta(1, 1) per action. Posterior update:

$$\alpha_a \leftarrow \alpha_a + \mathbb{1}[s'.\texttt{activity} = \texttt{active}], \quad \beta_a \leftarrow \beta_a + \mathbb{1}[s'.\texttt{activity} = \texttt{sedentary}]$$

Action selected by sampling $\theta_a \sim \text{Beta}(\alpha_a, \beta_a)$ and choosing $a^* = \arg\max_a \theta_a$.

2.2 Epsilon-Greedy

$\epsilon = 0.1$. With probability ϵ, select uniformly at random; otherwise select $\arg\max_a Q(a)$. Q-values updated via incremental average:

$$Q(a) \leftarrow Q(a) + \frac{r_t - Q(a)}{n_a}$$

2.3 Random Baseline

Uniform random action selection. No learning. Serves as lower bound.

3 Initial Results

All agents run on the same config (`config/mvp.yaml`), seed 42, 450 timesteps.

Agent	Total Reward	Mean Reward/Step
Thompson Sampling	158.0	0.3511
Epsilon-Greedy ($\epsilon = 0.1$)	163.0	0.3622
Random	138.0	0.3067

Table 2: Results on `config/mvp.yaml` (seed 42, 450 timesteps). TS and ϵ-greedy outperform random despite Reactance effect.

4 Known Limitations

- Agents are state-blind bandits, not state-aware RL. State-aware agents are Phase 2.
- Single-timescale reward only. Multi-timescale (immediate + 3-week delayed) is Phase 2.
- No user response model. The `RuleBasedResponse` module exists as a stub. Burden accumulation and 4 archetypes are Phase 2.

- Transition matrix is fixed and known. Learned transitions from real data are Phase 2.
- Transition probabilities are placeholder values informed by Reactance theory. Calibration against real data is Phase 2.